Systems of Linear Equations

ANSWER KEY

Section 1: Augmented Matrices and Row Echelon Form (REF)

1 Which matrix represents a valid Reduced Row Echelon Form (RREF)?

a. $[[1, 0, 3], [0, 1, -2], [0, 0, 0]]$ ✓

b. $[[1, 2, 3], [0, 1, -2], [0, 0, 0]]$

c. $[[2, 0, 0], [0, 1, 0], [0, 0, 1]]$

d. $[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$

In RREF, each leading entry (pivot) is 1, is the only non-zero entry in its column, and appears to the right of pivots in previous rows, with zero rows at the bottom.

2 How many pivots does a 3×5 matrix have at most?

a. 3 ✓

b. 5

c. 2

d. 15

Because each pivot must reside in a distinct row and a distinct column, the maximum number of pivots is $\min(m, n) = \min(3, 5) = 3$.

3 An augmented matrix has a row of the form $[0, 0, \ldots, 0 \mid b]$ with $b \neq 0$. What does this establish about the linear system?

a. **The system is inconsistent (has no solution)** ✓

b. The system has a unique solution

c. The system has infinitely many solutions

d. The determinant is equal to b

The row represents the contradictory equation $0 = b$. Any system with such a pivot in the augmented column is inconsistent.

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Section 2: Solution Sets and Free Variables

4 A consistent system has 5 variables and 3 pivot columns in its coefficient matrix. How many free variables does the system have?

a. 2 ✓

b. 3

c. 5

d. 0

The number of free variables equals total variables minus pivot columns: $5 - 3 = 2$ free variables, yielding a 2-dimensional plane of solutions.

5 When does a linear system $A\mathbf{x} = \mathbf{b}$ possess a unique solution?

a. **When the system is consistent and every column of A contains a pivot (no free variables)** ✓

b. Whenever the number of equations equals the number of variables

c. Whenever $\mathbf{b} \neq \mathbf{0}$

d. When there is at least one free variable

A consistent system has a unique solution if and only if every variable is a basic variable, meaning there are zero free variables.

6 If a system is consistent and has at least one free variable, how many solutions exist over $\mathbb{R}$?

a. **Infinitely many** ✓

b. Exactly one

c. None

d. Finite number equal to the number of free variables

Because free variables can take any real value, any consistent system with at least one free variable has infinitely many solutions.

Section 3: Homogeneous Systems and Null Space

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7 What is always true about the homogeneous linear system $A\mathbf{x} = \mathbf{0}$?

a. **It is always consistent, because $\mathbf{x} = \mathbf{0}$ is always a solution (the trivial solution) ✓**

b. It has no solutions if $\det(A) = 0$

c. It always has infinitely many solutions

d. It only has solutions when A is square

Setting all variables to zero ($x_1 = \ldots = x_n = 0$) always satisfies $A\mathbf{0} = \mathbf{0}$, so homogeneous systems are never inconsistent.

8 Under what condition does the homogeneous system $A\mathbf{x} = \mathbf{0}$ have non-trivial solutions (solutions where $\mathbf{x} \neq \mathbf{0}$)?

a. **If and only if the system has at least one free variable ✓**

b. If and only if A has more rows than columns

c. If and only if all entries of A are positive

d. Whenever $\mathbf{b} = \mathbf{0}$

Non-trivial solutions exist if and only if there is at least one free variable (i.e. the columns of A are linearly dependent).

9 The set of all solutions to $A\mathbf{x} = \mathbf{0}$ forms a subspace of $\mathbb{R}^n$ called the:

a. **Null space (or kernel) of A ✓**

b. Column space of A

c. Row space of A

d. Span of $\mathbf{b}$

By definition, $\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$ is the null space of the matrix.

Section 4: Matrix Invertibility and Systems

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10 According to the Invertible Matrix Theorem, for a square $n \times n$ matrix A, which statement is equivalent to A being invertible?

a. **The equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$** ✓

b. The columns of A are linearly dependent

c. $\det(A) = 0$

d. The null space of A contains non-zero vectors

Invertibility is equivalent to A having n pivots, meaning $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$ for any $\mathbf{b}$.

11 If A is invertible and $A\mathbf{x} = \mathbf{b}$, how is the unique solution expressed algebraically?

a. $\mathbf{x} = A^{-1}\mathbf{b}$ ✓

b. $\mathbf{x} = \mathbf{b}A^{-1}$

c. $\mathbf{x} = \frac{\mathbf{b}}{A}$

d. $\mathbf{x} = A^{T}\mathbf{b}$

Multiply both sides on the left by A^{-1}: $A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b} \Rightarrow I\mathbf{x} = A^{-1}\mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$.

12 If A is a 4×4 matrix with rank 3, how many solutions does $A\mathbf{x} = \mathbf{0}$ have?

a. **Infinitely many** ✓

b. Exactly one

c. None

d. 3

By the Rank-Nullity Theorem, $\text{nullity}(A) = n - \text{rank}(A) = 4 - 3 = 1$. With a 1-dimensional null space, there are infinitely many solutions.

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